INCH-POUND
MIL-PRF-1/1165F
2 December 2008
SUPERSEDING
MIL-E-1/1165E
15 June 1981

PERFORMACE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6948

Inactive for new design
after 21 July 1997

This specification is approved for use by all Depart-
ments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein
shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Double triode, subminiature, high Mu
Outline 3-11 (EIA).
Base E8-10
Envelope T3
Cathode Coated unipotential

Base connections

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	2a	2g	H	2k	1k	H	1g	1a

ABSOLUTE- MAXIMUM RATINGS:

Parameter:	Ef	Eb	eb	Ec	Ehk	Rk/k	t
Unit:	V	Vdc	v	Vdc	V	Ohms	sec
Maximum	6.9	250	360	0,-55	200	----	---
Minimum	5.5	----	----	-----	-----	-----	----
Absolute Maximum for Accelerated Warmup:	9.5	----	----	----	150	----	10
Test conditions	6.3	100	---	0	0	1,500	---

ABSOLUTE RATINGS:

Parameter:	Cycles	Rg/g	Ik/k	Pp/a	TE	Alt	Radiation	
Unit:	----	Meg Ω	mAdc	W	°C	ft	NVT	NV
Maximum	---	1.0	10	0.5	250	Note 1	10^16	10^12
Minimum	----	-----	-------	------	-------	-------	------	
Absolute maximum for accelerated warmup:	50	-----	-----	-----	------	------	------	
Test conditions	-----	-----	-----	-----	------	------	-------	

GENERAL:

Qualification – Not Required
Reliable tube

AMSC N/A

FSC 5960

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TABLE I. Testing and inspection.

Inspection	MIL-STD-1311 Method	Notes	Conditions	Acceptance level 16/	Symbol	Limits		Unit
						Min	Max	
Radiation rating	-----	2	----	---	--	---	---	---
Conformance inspection, part 1								
Heater current	1301			0.4	If	330	370	mA
Heater-Cathode leakage	1336	3		0.4	Ihk	----	15.0	µAdc
Total grid current	1266	3, 4	Eh = 150 Vdc; Ec = -1.5 Vdc; Rk = 0; Rg = 1.0 Meg Ω	0.4	Ic	0	-0.3	µAdc
Electrode current (1) (anode)	1256	3		0.4	Ib	0.50	1.10	mAdc
Electrode current (2 (anode)	1256	3	Ec = -3.5 Vdc; Rk = 0	0.4	Ib	---	50	µAdc
Transconductance (1)	1306	3		0.4	Sm	1,350	1,950	µohms
Short and discontinuity detection	1201			0.4	----	----	----	----
Conformance inspection, part 2								
Insulation of electrodes	1211	3		2.5	R	500	----	MegΩ
Electrode current (1) (anode) (difference between sections)	1256			2.5	Ib	---	0.2	mAdc
Transconductance (2)	1306	3	Ef = 5.5 V	2.5	$\frac{\Delta Sm}{Ef}$	----	15	%
Grid emission	1266	3, 5	Ef = 7.5 V; Ec = - 3.5 Vdc; Rg = 1.0 MegΩ; Rk = 0	2.5	Ic	0	-0.5	µAdc
Audio Frequency noise	1246	6, 7	Ecal = 45 mVdc; Rg = 0.1 MegΩ Rk= 750 ohms; Rp = 0.2 MegΩ	2.5	EB	---	---	-----
Noise and microphonics (hum)	2201	3, 8	Ef = 6.3 Vac at 400 Hz; Eb = Ec = 0; Rk/k = 5,000 ohms	2.5	----	-----	15	mvpp
Operation time	----	3, 16		6.5	t	---- --	20	sec
Amplification factor	1316	3		6.5	Mu	60	85	---
AC amplification	1321	3	Ebb = 100 Vdc; Ecc = 0; Rk =0; Esig = 0.2 Vac	6.5	Ep	7.5	---	Vdc

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TABLE I. Testing and inspection. Continued

Inspection	MIL-STD-1311 Method	Notes	Conditions	Acceptance level 16/	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 2 - Continued								
Direct- interelectrode capacitance	1331	3	No Shield	6.5	Cgp	0.55	0.95	pF
		3	No Shield		Cin	1.2	2.0	PF
			No Shield (Triode 1)		Cout	0.15	0.25	pF
			No Shield (Triode 2)		Cout	0.19	0.31	pF
			No Shield		Cgg	----	0.014	pF
			No Shield		Cpp	----	0.86	pF
High- frequency vibration	1031	3	Rp = 10,000 ohms	1.0	Ep	-----	10	mVac
White noise	----	3, 9	Peak accel = 15G;	2.5	ep	-----	60	mvpp
			Rp = 10,000 ohms	2.5	Ep	-----	10	mV
Lead fatigue	1116			2.5	----	-----	----	-----
Shock	1041	10	Accel = 750 G	15	----	-----	-----	-----
Sweep-frequency – vibration fatigue-test end points	-------		F = 30 to 3,000 Hz; t = 3 minutes	6.5	------	-----	-------	-----
	1031	3	Rp = 10,000 ohms	------	Ep	--------	50	mVac
High-frequency vibration	1336	3		-----	Ink	-------	15	µAdc
Heater-cathode leakage Transconductance (1) Change of individua tubes	1306	3		-----	ΔSm T	-----	20	%
Envelope strain	2126			4.0	------	----	-----	----
Permanence of marking	1105			----	-----	------	-----	------
Conformance inspection, part 3								
Heater-cycling life (1)	1506		Ef = 7.0V, 1 minute on, 4 minute off; Ehk = 140 Vac; Eb= Ec = 0	2.5	-------	------	----	-----
Heater-cycling life (2)	1506		Ef = 8.5 V, 10 seconds on, 4 minutes (min) off; 300 cycle; Ehk = 150 Vdc; Eb = Ec = 0	10	------	-----	------	------
Heater-cycling life (1) and (2) test end points:	-----							
Heater-cathode leakage	1336	3			Ihk	-----	15	µAdc
Insulation of electrodes	1211	3			R	50	----	MegΩ

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TABLE I. Testing and inspection. Continued

Inspection	MIL-STD-1311 Method	Notes	Conditions	Acceptance level 17	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 3 - Continued								
Stability life	1516		Ebb = 250 Vdc; Rk/k = 270 ohms; Ehk = 200 Vdc; Rg/g = 1.0 MegΩ Rp/a = 30,000 ohms; TA = room; Rhk = 5,000 ohms	-----	-----			
Stability life-test end point:	------							
Transcoductance (1). Change of individual tubes	1306	3			ΔSm t	-----	15	%
				------	-------	-----	----	-----
Intermittent life	1501	12, 16	Ebb = 250 Vdc; Rk/k = 270 ohms; Ekk = 200 Vdc; Rg/g = 1.0 MegΩ; Rp/a = 30,000 ohms; Rnk = 5,000 ohms; TE = 250 °C (min);					
Intermittent life-test end points: (200 hours);	-------	13						
Total grid current	1266	3	Eb = 150 Vdc; Ec = -1.5 Vdcl Rk = 0; Rg= 1.0 MegΩ		Ic If	0 330	-1.0 380	µAdc mA
Heater current								
Transconductance (1), Change of individual tubes	1301 1306	 3			ΔSm t	----	25	%
Transconductance (2)	1306	3	Ef = 5.5 V		ΔSm Ef	------	20	%
Heater-cathod leakage	1336	3						
Insulation of electrodes	1211	3			Ihk	-----	15	µAdc
Tansconductance (1), average change	1306				R Avg ΔSm t	200 -----	---- 15	Meg Ω %

NOTES:

1. See “Reduced pressure (altitude) rating,” and altitude, maximum peak voltage in the basic document.
2. This test shall be conducted in a suitable reactor with the required intensity and dosage that is not less than 10 percent of the specified rating. Before and after irradiation , the tubes under test (TUT) shall be within the specified limits for all electrical tests listed herein under quality conformance inspection, part 1.
3. Test each unit separately.
4. This test shall be performed at the conclusion of the holding period.

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5. Prior to this test, the tubes under test (TUT) shall be preheated for 5 minutes at the conditions specified below. Test with in 3 seconds after preheating. The 3-minute test shall not be permitted. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef	Eb	Ec	Rk/k	Rg/g
V	Vdc	Vdc	Ohms	MegΩ
7.5	250	0	1,300	1.0

6. Tie 1k to 2k; 1 g to 2g; and 1a to 2a.

7. The rejection level shall be set at the VU meter reading obtained during calibration.

8. Maximum total distortion of the filament supple voltage shall be 5 percent. The frequency response of the peak-to-peak measuring device from 20 to 5,000 Hz shall be within 0.5 dB of its response at 400Hz. Ground leads except those for the cathode and heater lead No. 3. Measure hum voltage across specified Rk.

9. The tube shall be rigidly mounted on at able vibrating so that the instantaneous values of acceleration shall constitute approximately a " white noise" spectrum which is free from discontinuities from 100 to 5,000 Hz and such that the rms value of acceleration for frequencies outside this band shall constitute no more than 5 percent of the total rms acceleration. The spectrum of instantaneous acceleration shall be such that each octave of bandwidth delivers 2.3 ± 0.2 G's rms acceleration. With this the case the rms value of acceleration for any bandwidth within the spectrum is equal to:

$$GRMGrms = 2.3\sqrt{3.32\log_{10}(F2/F1)}$$

Where F2 and F1 are the upper and lower frequencies respectively of the band under consideration. The degree of clipping of the clipping of the peak accelerations shall be such that the peak value of accelerations shall be such that the peak value of acceleration is at least 15 G's. Half the tubes in the sample shall be vibrated in position X, the other half in position Y. The voltage (ep) produced across the resistor (Rp) as a result of vibration shall be coupled through a compensation amplifier to a low-pass filter. The compensating amplifier shall have a high-input impedance (250 K ohms or more) and shall be adjusted to compensate for any insertion losses in the filter. The combined frequency response of amplifier and filter shall be flat within± 0.5 dB from 50 to 8,000 Hz, shall be down no more than 5 dB at 10,000 Hz and 20 Hz, and down at least 30 dB at 13,000 Hz. For reading peak-to-peak value of output voltage, the filter output shall be fed directly to the input of a Ballantine Model 305 peak-to-peak Electronic Voltmeter, or equal, while the rms value shall be measured with the Hewlett Packard Model 400 c, or equal. The impedance of the anode and screen voltage supplies shall not exceed that of a 40 µF capacitor at 10 Hz.

10. A grid resistor of 0.1 MegΩ shall be added except when a thyratron-type short indicator is used.

11. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When on lot has passed, the 12-month rule shall apply. In the event of lot failure, the lat shall be rejected and the succeeding lots shall be subjected to this test until a lot passes.

12. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurements (MIL-STD-1311 method 1226), will be satisfied if a TUT having bogey Ib (± 5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

13. The allowable defectives per characteristic shall not be greater than 1 for first sample and shall not be greater than 3 for combined samples. Total defective shall be 2 and 5, respectively.

14. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

15. Insert a cold tube into the test socket having all electrode current (1) (anode) conditions applied and record Ib continuously for 3 minutes. Ib shall reach 85 percent of the 3 minutes figure within the time indicated.

16. The life-test sample shall consist of 20 tubes per lot and not tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be elected from the lot. Acceptance shall then be based on the second samples and there shall be zero tube failures.

17. This specification sheet utilizes an accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

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Referenced documents. In addition to MIL-PRF-1, this document references the following:
MIL-STD-1311

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – CR
Navy – EC
Air Force – 85

Preparing activity:
DLA- CC
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Review activities:
Navy – AS, CG, MC, OS, SH
Air Force – 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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